

Actual Exposure and Low Pair Codegree in an Activated Fixed-Shift Residual Gram: Row Singletons, Forest-Exact Lower Frames, and Cycle-Holonomy Obstructions

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Abstract

The activated-range Gram from the preceding fixed-shift analysis is exact but abstract: its smallest eigenvalue depends on the actual row columns after residual grouping. We identify their literal support geometry. On one declared source system, a fixed physical row contributes at most one terminal atom to a given output fiber. Hence every active normalized row column has diagonal energy exactly one; all loss is cross-row collision. Summing the factor-exchange and CRT geometry of an earlier local analysis yields a new uniform row-pair codegree law. If the root cofactor is in $[R, 2R]$, two distinct rows share at most $X^{o(1)}(1 + J/R)$ reduced output cells, and therefore only $X^{o(1)}$ cells on the critical block $R \asymp J$, even though one output fiber can still have polynomial row degree. We then connect this sparse incidence to the lower spectrum. Private-output energy gives a sharp lower frame; an optimal phase-blind comparison matrix gives weighted diagonal dominance; and forest Gram graphs admit an exact formula because all edge phases can be gauged away. Unique-matching minors give phase-uniform injectivity, whereas ordinary Hall matching does not. A cyclic model with identical support, degrees, codegrees, and magnitudes is singular or nonsingular according only to one holonomy phase. Thus low pair codegree is a genuine structural advance, but it reaches a polynomial physical lower bound only after row-energy diffuseness or a global expansion/holonomy estimate is proved. No such endpoint estimate, parity improvement, or prime-pair conclusion is claimed.

Keywords: fixed shift; residual fiber; pair codegree; actual support; lower frame; unique matching; cycle holonomy.

Convention. Arithmetic specializations use one fixed nonzero shift h_0 , literal physical coefficients, actual nonzero support, and raw counting-measure atomic norms. The shift is not specialized to 2. Inner products are conjugate linear in the first variable.

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1 Introduction

TPC-65 reduced the represented-reference question to the smallest eigenvalue of an actual row Gram on the terminal range [5]. Its literal matrix entries are residual-fiber correlations. That result closed the carrier and normalization problem, but deliberately left two arithmetic questions: how often are two physical rows exposed to the same output, and can those collisions create a small singular direction?

This paper answers the first question at the critical cofactor scale and develops exact support-level tests for the second. The starting observation is simple but decisive. For the physical key

$$q(m, r, j, \gamma) = (\gamma, d_m r, j),$$

fixing m and q fixes r, j , and γ , and the terminal prime is then fixed by $pr - mj = h_0$. Thus a row never coherently sums two of its own terminal atoms inside one output coordinate. The normalized physical row Gram has unit diagonal. Its off-diagonal graph is the entire problem.

The arithmetic result uses, but does not rebrand, the factor-exchange and CRT identities of TPC-57 [3]. Those identities treat one fixed exchange pattern. Here the rows are fixed and all admissible exchange factors are summed. A divisor-thinning argument leaves only $X^{o(1)}$ factors, and the CRT modulus is at least the root cofactor R . Consequently the reduced pair codegree is

$$C_{m,n}^{\text{red}}(R) \leq X^{o(1)} \left(1 + \frac{J}{R}\right).$$

At $R \asymp J$ this is subpower uniformly in the distinct row pair. This is not the older fiber-degree estimate: the same skeleton can have high right degree and low left pair codegree.

Pairwise sparsity is not yet a lower frame. We therefore prove a hierarchy of exact finite-dimensional results. Private output energy gives a sharp positive lower bound. The absolute overlap matrix gives the optimal phase-blind comparison bound and its weighted Collatz–Wielandt form. On forests, all edge phases are gauge, giving an exact spectral-radius formula and a sharp star law. At the support level, Hall deficiency is a universal rank obstruction, while a unique-matching minor is a universal injectivity certificate. Ordinary Hall matching lies strictly between them: it gives only generic full rank and does not settle the one literal coefficient array.

The boundary is a cycle. A degree-two, codegree-one support pattern with fixed entry magnitudes can be singular or nonsingular depending on one cycle holonomy. This proves that exposure, bounded degrees, bounded codegrees, Hall matching, and magnitude data do not determine the physical lower frame. The next arithmetic gate is therefore global: combine the proved reduced-cell pair thinning with actual full-output diffuseness and expansion, or control the remaining holonomy. It is not legitimate to infer such control from support counts.

Claim levels

The support and matrix theorems are L0. The row-singleton dictionary, factor-exchange summation, CRT spacing, and divisor bound are unconditional L1 statements on one literal fixed- h_0 carrier. A positive exposure lower bound, polynomial diffuseness, global smallest-singular-value estimate, fixed- $h_0 = 2$ parity estimate, or twin-prime conclusion would be L2 and is not proved here.

2 Literal row singletons and the unit diagonal

Fix one declared source system, one cofactor block $R \leq r \leq 2R$, and one external block label until its explicit erasure. A terminal atom is

$$u = (m, r, j, \gamma), \quad m = d_m \ell_m,$$

with terminal prime

$$p_u = \frac{mj + h_0}{r}$$

and physical residual key

$$q(u) = (\gamma, s = d_m r, j). \tag{1}$$

Let $\mathcal{U}_X(R)$ be the actual nonzero terminal atoms in this scope. Write their literal coefficients as $a_u \neq 0$. For every physical row m and output y , define

$$b_{y,m} := \sum_{\substack{u \in \mathcal{U}_X(R) \\ m(u)=m, q(u)=y}} a_u. \quad (2)$$

The physical row synthesis is

$$(B_X z)_y = \sum_m b_{y,m} z_m.$$

Lemma 2.1 (Row-output singleton). *For fixed m and y , the sum in (2) contains at most one terminal atom. Consequently*

$$b_{y,m} = a_u \text{ for the unique such atom, or } b_{y,m} = 0. \quad (3)$$

Proof. Write $y = (\gamma, s, j)$. The row fixes d_m , and the output fixes $r = s/d_m$, j , and γ . The terminal equation then fixes $p = (mj + h_0)/r$. The external block label has been retained, so there is no second copy from another declared block. Thus the atom is unique. $\square$

Let

$$T_m(R) := \sum_{\substack{u \in \mathcal{U}_X(R) \\ m(u)=m}} |a_u|^2$$

be the terminal energy in row m , and retain only active rows $\mathcal{M}_X^+(R) = \{m : T_m(R) > 0\}$. Define

$$v_m(y) := \frac{b_{y,m}}{\sqrt{T_m(R)}}, \quad K_X(m, n) := \langle v_m, v_n \rangle.$$

Theorem 2.2 (Exact unit-diagonal residual Gram). *If the active row space is nonempty, then on that space*

$$K_X(m, m) = 1 \quad (m \in \mathcal{M}_X^+(R)). \quad (4)$$

The represented-to-terminal lower constant on this block is

$$\beta_X(R) = \lambda_{\min}(K_X). \quad (5)$$

Thus every loss from the residual grouping occurs in cross-row correlations; there is no within-row coherent loss.

Proof. By lemma 2.1, distinct atoms from a fixed row occupy distinct output coordinates. Hence

$$\sum_y |b_{y,m}|^2 = \sum_{u: m(u)=m} |a_u|^2 = T_m(R),$$

which proves (4). Equation (5) is the activated-range Gram identity of Wang [5] in this literal basis. $\square$

Remark 2.3 (Scope). The theorem is not an ambient-incidence assertion. Zero physical coefficients have already been removed, the terminal prime equation is imposed, the block label is retained, and the norm is the raw atomic norm. The inherited factorization $m = d_m \ell_m$ is fixed on each physical row, and $u = (m, r, j, \gamma)$ is the declared atomic coordinate: no additional label has been silently quotiented out in this scope. If multiple block labels are coherently erased, the resulting cross-block map is a later reassembly gate.

2.1 Actual support and collision data

Define

$$S_m := \{y : v_m(y) \neq 0\}, \quad d_y := |\{m : y \in S_m\}|,$$

and for distinct rows

$$c_{m,n} := |S_m \cap S_n|.$$

These are actual supports after terminal restriction. They are not potential congruence classes. Double counting gives

$$\sum_m |S_m| = \sum_y d_y, \tag{6}$$

$$\sum_{m < n} c_{m,n} = \sum_y \binom{d_y}{2}. \tag{7}$$

Moreover, if $d_y \leq D_R$ for every output, then the collision load rooted at one row satisfies

$$\sum_{n \neq m} c_{m,n} = \sum_{y \in S_m} (d_y - 1) \leq (D_R - 1)|S_m|. \tag{8}$$

These identities distinguish right degree d_y from left pair codegree $c_{m,n}$; neither determines the other.

3 Actual exposure by cofactor–orbit cells

The row-singleton theorem identifies outputs with literal atoms, but it does not prove that a row contains any such atom. We record the exact exposure decomposition and its coefficient-blind upper bound.

For an active row m , a cofactor $r \in [R, 2R]$, and an orbit time j , put

$$\Gamma_m(r, j) := \{\gamma : (m, r, j, \gamma) \in \mathcal{U}_X(R)\}$$

and

$$E_m(r, j) := \sum_{\gamma \in \Gamma_m(r, j)} |a_{m,r,j,\gamma}|^2. \tag{9}$$

Then the exposure and energy disintegrate exactly:

$$S_m = \bigsqcup_{\substack{r \in [R, 2R] \\ j}} \{(\gamma, d_m r, j) : \gamma \in \Gamma_m(r, j)\}, \tag{10}$$

$$T_m(R) = \sum_{r,j} E_m(r, j). \tag{11}$$

Proposition 3.1 (One-row orbit spacing). *For sufficiently large X , fix a literal row $m = d_m \ell_m$ and a retained cofactor r . If one solution of*

$$rp - mj = h_0$$

exists, all integral solutions are

$$j = j_0 + rt, \quad p = p_0 + mt, \quad t \in \mathbb{Z}. \tag{12}$$

Consequently an orbit interval of length $O(J)$ contains at most

$$1 + O\left(\frac{J}{r}\right) \leq 1 + O\left(\frac{J}{R}\right) \tag{13}$$

candidate j -values for this (m, r) .

Proof. On the inherited terminal support, $(d_m, r) = 1$. If the source prime ℓ_m divided r , then it would divide both rp and $d_m\ell_m j$, hence h_0 , impossible once $\ell_m > |h_0|$. Thus $(m, r) = 1$. Differences of two solutions obey $r\Delta p = m\Delta j$, yielding (12). The spacing of j is r , which gives (13). $\square$

Primality of p , the mask, the Mellin profile, and every other literal support condition only delete these candidates. Therefore (13) is an upper census. It proves neither nonemptiness nor a lower bound for $T_m(R)$. Even a large candidate count does not imply energy diffuseness among the surviving cells.

4 Divisor-thin reduced codegree of a fixed row pair

The determinant-ladder bound of TPC-55 controls how many rows can enter one residual output [2]. That is a right-degree estimate. The present section proves a different statement: after two source rows are fixed, the number of reduced residual cells in which they can meet is much smaller on a critical cofactor block. The proof sums the factor-exchange and simultaneous-CRT description of TPC-57 over every exchange cofactor that can occur [3].

4.1 Reduced collision cells

Work on one declared source system and one fixed direct residue/orbit slice of the literal terminal carrier. Fix $h_0 \neq 0$. Source rows have their inherited unique factorizations

$$m_i = d_i \ell_i, \quad d_i \in [D, 2D], \quad \ell_i \in [L, 2L] \cap \mathbb{P}, \quad i = 1, 2, \quad (14)$$

where the d_i are squarefree. A terminal atom on row m_i has cofactor r_i , terminal prime p_i , and satisfies

$$r_i p_i - m_i j = h_0. \quad (15)$$

The literal equality support also has

$$d_i, r_i, d_i r_i \text{ squarefree}, \quad (d_i, r_i) = 1.$$

Every prime, residue, mask, profile, and endpoint restriction remains part of the carrier. Let $\mathcal{J}_X$ denote the retained orbit interval and fix $C_J \geq 1$, independent of the rows and X , such that

$$\text{diam}(\mathcal{J}_X) \leq C_J J.$$

For $R > 0$ and two distinct rows $m_1 \neq m_2$, define $\mathcal{C}_{m_1, m_2}^{\text{red}}(R)$ to be the set of pairs (s, j) for which there are opposite labels γ_1, γ_2 and two literal terminal atoms

$$u_i = (m_i, r_i, j, \gamma_i), \quad d_i r_i = s, \quad i = 1, 2,$$

with $R \leq r_1 \leq 2R$. Put

$$c_{m_1, m_2}^{\text{red}}(R) := \left| \mathcal{C}_{m_1, m_2}^{\text{red}}(R) \right|. \quad (16)$$

Thus each row must be genuinely exposed over the reduced cell, but the two opposite labels need not agree and are not counted with multiplicity. This stronger reduced codegree contains the set of arithmetic cells that can contribute to the row-pair Gram entry. It is not the cardinality of the intersection of the two full (γ, s, j) -supports. The definition is rooted at m_1 and may be asymmetric. The symmetric intersection in which both cofactors lie in $[R, 2R]$ is a subset of this rooted set, in either ordering.

4.2 Exact factor exchange and the simultaneous orbit class

The following lemma records the fixed-row consequences of the factor-exchange normal form. It is included to make the subsequent summation transparent.

Lemma 4.1 (Fixed-row exchange equations and determinant). *Let $(s, j) \in \mathcal{C}_{m_1, m_2}^{\text{red}}(R)$. There are unique positive, pairwise coprime, squarefree integers a, b, c, e such that*

$$d_1 = ab, \quad d_2 = ac, \quad r_1 = ce, \quad r_2 = be, \quad s = abce. \quad (17)$$

For fixed rows, $a = (d_1, d_2)$, $b = d_1/a$, and $c = d_2/a$ are fixed; only e can vary. The two terminal equations are

$$ce p_1 - abl_1 j = h_0, \quad (18)$$

$$be p_2 - acl_2 j = h_0, \quad (19)$$

and imply

$$e(cp_1 - bp_2) = aj(bl_1 - cl_2), \quad (20)$$

$$\frac{e}{(e, h_0)} \mid bl_1 - cl_2. \quad (21)$$

Moreover,

$$bl_1 - cl_2 \neq 0. \quad (22)$$

Proof. The unique squarefree factor exchange is the one from TPC-57: the prime factors of s are partitioned according to whether they belong to both d_i , only d_1 , only d_2 , or both r_i . Substitution in the two copies of (15) gives (18) and (19); subtracting them gives (20).

On the literal carrier, $(m_i, r_i) = 1$. Reducing the two terminal equations modulo r_i therefore gives

$$j \equiv -h_0(abl_1)^{-1} \pmod{ce}, \quad j \equiv -h_0(acl_2)^{-1} \pmod{be}.$$

Compatibility modulo the common factor e is equivalent to $e \mid h_0(bl_1 - cl_2)$, which is (21). Finally, equality in (22) would give $m_1 = abl_1 = acl_2 = m_2$, contrary to the hypothesis. $\square$

Proposition 4.2 (Exact CRT class and two terminal-prime forms). *Fix the two distinct rows and an admissible exchange cofactor e arising from lemma 4.1; in particular, e is squarefree, $(e, abc) = 1$, and it satisfies (21). The two orbit congruences are compatible and determine one residue class*

$$j \equiv j_e \pmod{bce}. \quad (23)$$

Consequently an orbit interval of diameter at most $C_J J$ contains at most

$$1 + \left\lfloor \frac{C_J J}{bce} \right\rfloor \quad (24)$$

simultaneously integral times.

Writing $j = j_e + bce q$, the two terminal candidates are

$$p_1(q) = p_{1,e} + ab^2 l_1 q, \quad p_2(q) = p_{2,e} + ac^2 l_2 q. \quad (25)$$

They are nonproportional: their cross determinant is

$$(ac^2 l_2) p_{1,e} - (ab^2 l_1) p_{2,e} = -\frac{ah_0}{e} (bl_1 - cl_2) \neq 0. \quad (26)$$

Proof. The two moduli are ce and be . Since b, c, e are pairwise coprime, their gcd and least common multiple are respectively e and bce . The compatibility condition is precisely (21), so the Chinese remainder theorem gives (23). The spacing bound gives (24).

Substituting $j = j_e + bce q$ in (18) and (19) gives (25). At $q = 0$, multiply the first terminal equation by cl_2 , the second by bl_1 , and subtract. This gives

$$e(c^2 l_2 p_{1,e} - b^2 l_1 p_{2,e}) = -h_0(bl_1 - cl_2).$$

Multiplication by a , followed by (22), proves (26). $\square$

4.3 Summing every exchange cofactor

Theorem 4.3 (Divisor-thin rooted fixed-row-pair codegree). *Assume that, for one fixed $A > 0$, all row and orbit labels are at most X^A , while $h_0 \neq 0$ is fixed. Uniformly over distinct literal source rows,*

$$c_{m_1, m_2}^{\text{red}}(R) \leq \sum_{\substack{e \geq 1 \text{ squarefree} \\ (e, abc)=1 \\ e/(e, h_0) | b\ell_1 - c\ell_2 \\ R \leq ce \leq 2R}} \left(1 + \left\lfloor \frac{C_J J}{bce} \right\rfloor\right) \quad (27)$$

$$\ll_{h_0, C_J} X^{o(1)} \left(1 + \frac{J}{R}\right). \quad (28)$$

The $o(1)$ is uniform over the declared row pair.

Proof. For fixed rows, a, b, c, ℓ_1, ℓ_2 are fixed. Put

$$N_{12} := b\ell_1 - c\ell_2 \neq 0.$$

For every admissible e , let $\delta = (e, h_0)$. Since e is squarefree, δ is one of the $O_{h_0}(1)$ divisors of $\text{rad } |h_0|$, while

$$e/\delta \mid N_{12}$$

by (21). The map $e \mapsto (\delta, e/\delta)$ is injective. Hence

$$\#\{e : e \text{ squarefree}, e/(e, h_0) \mid N_{12}\} \leq \tau(\text{rad } |h_0|)\tau(|N_{12}|) = X^{o(1)}, \quad (29)$$

uniformly, by the standard divisor bound and the polynomial size of N_{12} .

For each e , [proposition 4.2](#) gives at most (24) possible times. Every literal primality, residue, mask, profile, and endpoint condition only deletes such times, proving (27). On the root block, $ce = r_1 \geq R$, and therefore

$$bce = br_1 \geq R.$$

Each summand in (27) is consequently $O_{C_J}(1 + J/R)$. Combining this with (29) proves (28). $\square$

Corollary 4.4 (Critical-block pair linearity). *If $R/J = X^{o(1)}$ and $J/R = X^{o(1)}$, in particular if*

$$R \asymp J = X^{133/400 + o(1)},$$

then, uniformly over distinct active rows,

$$\boxed{c_{m_1, m_2}^{\text{red}}(R) = X^{o(1)}}. \quad (30)$$

Proof. Insert $J/R = X^{o(1)}$ in (28). $\square$

4.4 Asymmetry with the right-fiber degree

The low pair codegree does not say that one residual output has low multiplicity. TPC-55 proves, on a cofactor range comparable to R ,

$$\Delta_R^{\text{fib}} := \max_y \#\{u : q(u) = y, r_u \asymp R\} \ll X^{o(1)} \left(1 + \frac{L}{R}\right). \quad (31)$$

The same bound applies to the reduced cell (s, j) obtained by forgetting γ : before the opposite-label masks are imposed, the candidate rows are already the source/terminal-prime solutions of the fixed- (s, j, d) determinant ladder, and allowing the existence of some γ does not create

an additional arithmetic solution. Opposite labels only decorate or delete such a row. The constant-factor saturation $r_u \in [R/2, 4R]$ changes no exponent. At the inherited endpoint,

$$L = X^{99979/210000+o(1)}, \quad R \asymp J = X^{133/400+o(1)},$$

so

$$\Delta_R^{\text{fib}} \leq X^{15077/105000+o(1)}, \quad \max_{m_1 \neq m_2} c_{m_1, m_2}^{\text{red}}(R) = X^{o(1)}. \quad (32)$$

Thus the literal reduced incidence may have polynomially large right degree while remaining subpower-linear from the viewpoint of every fixed left row pair.

4.5 A four-cycle consequence

Let $\mathcal{G}_R^{\text{red}}$ be the bipartite graph whose left vertices are source rows and whose right vertices are reduced cells (s, j) ; join m to (s, j) when a literal terminal atom on row m , with $r \in [R, 2R]$, occurs over that cell for at least one γ . Write

$$I_m(R) := \{(s, j) : m \sim (s, j)\}, \quad H_m(R) := |I_m(R)|,$$

and let $d_R(s, j)$ be the right degree. Put

$$C_R := \max_{m \neq n} |I_m(R) \cap I_n(R)|, \quad \Delta_R := \max_{(s, j)} d_R(s, j).$$

Corollary 4.5 (Rooted four-cycle bound). *The number $\mathfrak{C}_4(m; R)$ of unoriented four-cycles of $\mathcal{G}_R^{\text{red}}$ containing a fixed left vertex m satisfies*

$$\mathfrak{C}_4(m; R) = \sum_{n \neq m} \binom{|I_m(R) \cap I_n(R)|}{2} \quad (33)$$

$$\leq \frac{\max\{C_R - 1, 0\}}{2} \sum_{(s, j) \in I_m(R)} (d_R(s, j) - 1) \quad (34)$$

$$\leq \frac{\max\{C_R - 1, 0\}}{2} H_m(R) (\Delta_R - 1). \quad (35)$$

At $R \asymp J$,

$$\mathfrak{C}_4(m; R) \leq H_m(R) X^{15077/105000+o(1)}. \quad (36)$$

Proof. A four-cycle through m is uniquely a second row $n \neq m$ together with two distinct common reduced cells, proving (33). For $0 \leq c \leq C_R$,

$$\binom{c}{2} \leq \frac{\max\{C_R - 1, 0\}}{2} c.$$

Moreover, double-counting the ordered pairs consisting of a cell incident to m and another row incident to that cell gives

$$\sum_{n \neq m} |I_m(R) \cap I_n(R)| = \sum_{(s, j) \in I_m(R)} (d_R(s, j) - 1).$$

These two identities prove (34); the maximal right-degree bound gives (35). Finally use [corollary 4.4](#) and (32). $\square$

Remark 4.6 (Literal scope and the γ -boundary). Every result in this section is L1: it is an unconditional upper statement on one literal fixed- h_0 terminal carrier, using the actual squarefree

factor exchange and the actual orbit congruences. The argument does not enlarge the carrier; omitted primality, mask, profile, residue, or endpoint tests can only reduce the counted set.

The reduction from (γ, s, j) to (s, j) is deliberate. The raw full-support intersection is

$$\sum_{(s,j) \in \mathcal{C}_{m_1, m_2}^{\text{red}}(R)} \#\{\gamma : b_{(\gamma, s, j), m_1} b_{(\gamma, s, j), m_2} \neq 0\},$$

and the inner multiplicity can be polynomially large. Therefore (30) is not a subpower theorem for raw γ -decorated support. Its role is to bound the number of arithmetic cells contributing to one row-pair Gram entry after the opposite-label coordinates are combined with their literal weights. No lower exposure, coefficient diffuseness, signed cancellation, smallest-eigenvalue estimate, parity improvement, or prime-pair conclusion is asserted.

5 From reduced codegree to the decorated Gram

The arithmetic codegree theorem counts reduced cells $c = (s, j)$, whereas a full physical output also carries γ . The γ -multiplicity must not be silently inserted into or removed from a support count.

For a reduced cell $c = (s, j)$, let Γ_c be the ambient set of opposite labels occurring over that cell. For a row m , define the decorated cell vector, extended by zero on unused labels,

$$V_{m,c} := (b_{(\gamma, s, j), m})_\gamma$$

in $\ell^2(\Gamma_c)$, with

$$E_m(c) := \|V_{m,c}\|^2.$$

Let $\mathcal{C}_{m,n}(R)$ be the common reduced cells of rows m, n in the represented block and put

$$C_{m,n}^{\text{red}}(R) := |\mathcal{C}_{m,n}(R)|.$$

This symmetric represented codegree is bounded by the stronger rooted codegree $c_{m,n}^{\text{red}}(R)$ of [section 4](#), in either row ordering. Then the normalized Gram entry has the exact disintegration

$$K_X(m, n) = \frac{1}{\sqrt{T_m(R)T_n(R)}} \sum_{c \in \mathcal{C}_{m,n}(R)} \langle V_{m,c}, V_{n,c} \rangle. \quad (37)$$

Whenever both cell energies are nonzero, define the opposite overlap

$$\omega_{m,n}(c) := \frac{|\langle V_{m,c}, V_{n,c} \rangle|}{\sqrt{E_m(c)E_n(c)}} \in [0, 1]. \quad (38)$$

Thus a reduced cell decorated by many γ 's is one Gram correlation, not automatically many independent phases.

Definition 5.1 (Cell diffuseness). For an active row, set

$$\rho_m(R) := \max_c \frac{E_m(c)}{T_m(R)} \in (0, 1]. \quad (39)$$

Proposition 5.2 (Codegree–diffuseness bridge). *For distinct active rows,*

$$|K_X(m, n)| \leq C_{m,n}^{\text{red}}(R) \sqrt{\rho_m(R)\rho_n(R)}. \quad (40)$$

On the critical block $R \asymp J$, the arithmetic codegree theorem therefore gives

$$|K_X(m, n)| \leq X^{o(1)} \sqrt{\rho_m(R)\rho_n(R)} \quad (41)$$

uniformly for distinct rows.

Proof. Each summand in (37) has magnitude at most

$$\sqrt{E_m(c)E_n(c)} \leq \sqrt{\rho_m T_m \rho_n T_n}.$$

There are at most $C_{m,n}^{\text{red}}(R)$ common reduced cells. Divide by $\sqrt{T_m T_n}$. $\square$

Corollary 5.3 (Concentration inverse statement). *If $|K_X(m, n)| \geq \eta > 0$, then*

$$\rho_m(R)\rho_n(R) \geq \frac{\eta^2}{(C_{m,n}^{\text{red}}(R))^2}.$$

At $R \asymp J$, a large off-diagonal correlation therefore forces both rows to place nonnegligible normalized energy in a small family of common arithmetic cells.

Neither TPC-64 nor TPC-65 proves a fixed-power upper bound for $\rho_m(R)$. Support cardinality cannot replace it: a row may have many nonzero cells and place almost all energy in one of them.

5.1 Why the residual Möbius sign is not cancellation

On the squarefree terminal support,

$$\mu(d_m)\mu(r) = \mu(s), \quad s = d_m r.$$

The source sign $\mu(d_m)$ and terminal sign $\mu(r)$ therefore combine in the literal coefficient into the output sign $\mu(s)$, common to every row contributing to (γ, s, j) . Multiplication of one output coordinate by a common unit scalar leaves every Gram entry unchanged. Thus the residual represented Gram cannot obtain a parity saving from these Möbius factors. The fixed-complexity mask analysis does not furnish small decorated overlap either: its canonical-cell theorem instead retains $1 - o(1)$ of the relevant product energy across the static mask, which is a different quantity from $\omega_{m,n}(c)$ [1]. Genuine signed two-affine cancellation belongs to the later native/reassembly map studied around TPC-61 [4], not to the signless residual skeleton isolated here.

6 Actual-support lower-frame certificates

This section begins after terminal activation and restriction to the actual activated range. Thus there is no semidefinite parent nuisance to eliminate and no shorted operator to recompute. The input is one finite matrix of literal represented columns, and the target is its smallest Gram eigenvalue. All inner products in this paper are conjugate-linear in the first variable.

6.1 Actual columns and diagonal normalization

Let $\mathcal{M}$ and $\mathcal{Y}$ be finite nonempty sets, and let

$$B : \mathbb{C}^{\mathcal{M}} \longrightarrow \ell^2(\mathcal{Y}), \quad (Bz)_y = \sum_{m \in \mathcal{M}} a_{y,m} z_m.$$

The m -th column is

$$g_m = (a_{y,m})_{y \in \mathcal{Y}}, \quad S_m := \{y \in \mathcal{Y} : a_{y,m} \neq 0\}.$$

Here S_m is the support after every coherent sum defining the entry $a_{y,m}$ has been taken. It is not the larger set of output coordinates which could have occurred before those sums. In the unit-diagonal fixed-shift application from the preceding sections,

$$a_{y,m} = \frac{b_{y,m}}{\sqrt{T_m(R)}}, \quad e_m = 1.$$

The slightly more general notation below keeps the diagonal-exposure gate visible and also applies after an additional declared output multiplier.

Define

$$G := B^* B, \quad G_{m,n} = \langle g_m, g_n \rangle = \sum_{y \in \mathcal{Y}} \overline{a_{y,m}} a_{y,n}, \quad \beta := \lambda_{\min}(G). \quad (42)$$

If one column is zero, then $\beta = 0$. We therefore assume below that

$$e_m := \|g_m\|^2 = \sum_{y \in S_m} |a_{y,m}|^2 > 0 \quad (m \in \mathcal{M}). \quad (43)$$

Put

$$D := \text{diag}(e_m), \quad v_m := \frac{g_m}{\sqrt{e_m}}, \quad C := D^{-1/2} G D^{-1/2}.$$

Then C is the Gram matrix of the unit columns v_m , and $C_{m,m} = 1$.

Proposition 6.1 (Diagonal exposure and normalized collision). *If*

$$e_{\min} := \min_m e_m, \quad e_{\max} := \max_m e_m, \quad \kappa := \lambda_{\min}(C),$$

then

$$\boxed{e_{\min} \kappa \leq \beta \leq e_{\max} \kappa.} \quad (44)$$

In particular, a lower-frame proof must control both the least diagonal exposure $e_{\min}$ and the normalized collision gap κ .

Proof. Since $G = D^{1/2} C D^{1/2}$, for every z ,

$$\langle z, Gz \rangle \geq \kappa \|D^{1/2} z\|^2 \geq e_{\min} \kappa \|z\|^2.$$

For the reverse inequality, let x be a unit eigenvector of C with eigenvalue κ , and set $z = D^{-1/2} x$. Then

$$\frac{\langle z, Gz \rangle}{\|z\|^2} = \frac{\kappa}{\langle x, D^{-1} x \rangle} \leq e_{\max} \kappa.$$

□

6.2 Private exposure

For $y \in \mathcal{Y}$, let

$$d_y := |\{m \in \mathcal{M} : y \in S_m\}|$$

be its actual output degree. The private support and private energy of column m are

$$P_m := \{y \in S_m : d_y = 1\}, \quad p_m := \sum_{y \in P_m} |a_{y,m}|^2. \quad (45)$$

Theorem 6.2 (Exact private-exposure certificate). *The represented Gram satisfies*

$$\boxed{G \succeq \text{diag}(p_m), \quad \beta \geq \min_{m \in \mathcal{M}} p_m.} \quad (46)$$

The coefficient one in this estimate is optimal.

Proof. Restrict B to the union of the private output coordinates. Because a private coordinate is occupied by exactly one column, the Gram of this restriction is $\text{diag}(p_m)$. The Gram of all remaining output coordinates is positive semidefinite. Their sum is G , proving the operator inequality and hence the eigenvalue bound.

For sharpness, take one private output for each column and no shared output. Then $G = \text{diag}(p_m)$, so equality holds. □

Corollary 6.3 (Private counts require atomic weight). *Suppose that numbers $\ell_m > 0$ satisfy*

$$|a_{y,m}|^2 \geq \ell_m \quad (y \in P_m).$$

Then

$$\beta \geq \min_m \ell_m |P_m|. \quad (47)$$

Without such a weight lower bound, the counts $|P_m|$, even when all are positive, imply no scale-uniform numerical lower bound.

Proof. The first assertion follows from $p_m \geq \ell_m |P_m|$ and (46). For the second, use disjoint supports and give every nonzero entry magnitude ε . The support pattern and all private counts remain fixed while $\beta = \varepsilon^2$ tends to zero. $\square$

6.3 The optimal phase-blind comparison matrix

For $m \neq n$, define the support-weight overlap

$$\Omega_{m,n} := \sum_{y \in S_m \cap S_n} |a_{y,m}| |a_{y,n}|, \quad \Omega_{m,m} := 0. \quad (48)$$

Thus Ω is a symmetric entrywise nonnegative matrix and

$$|G_{m,n}| \leq \Omega_{m,n}. \quad (49)$$

Its comparison matrix is

$$Q_\Omega := \text{diag}(e_m) - \Omega. \quad (50)$$

Theorem 6.4 (Comparison matrix and optimized weighted Gershgorin). *For every finite matrix B above,*

$$\boxed{\beta \geq \max\{0, \lambda_{\min}(Q_\Omega)\}}. \quad (51)$$

Moreover,

$$\boxed{\lambda_{\min}(Q_\Omega) = \sup_{t \in (0, \infty)^{\mathcal{M}}} \min_{m \in \mathcal{M}} \left(e_m - \sum_{n \neq m} \Omega_{m,n} \frac{t_n}{t_m} \right)}. \quad (52)$$

No irreducibility hypothesis is required. In the present symmetric setting, a positive maximizing vector can be assembled from the positive ground-state vectors of the connected components.

Proof. For $z \in \mathbb{C}^{\mathcal{M}}$, the triangle inequality gives

$$\begin{aligned} \langle z, Gz \rangle &= \sum_m e_m |z_m|^2 + 2 \operatorname{Re} \sum_{m < n} \overline{z_m} G_{m,n} z_n \\ &\geq \sum_m e_m |z_m|^2 - 2 \sum_{m < n} \Omega_{m,n} |z_m| |z_n| \\ &= \langle |z|, Q_\Omega |z| \rangle_{\mathbb{R}}. \end{aligned}$$

Rayleigh–Ritz gives $\lambda_{\min}(G) \geq \lambda_{\min}(Q_\Omega)$; positivity of G allows the right-hand side to be replaced by its positive part.

For the variational identity, choose $s \geq \max_m e_m$ and put

$$A := sI - Q_\Omega.$$

Then A is entrywise nonnegative and

$$\rho(A) = s - \lambda_{\min}(Q_\Omega).$$

The upper Collatz–Wielandt formula, valid for every finite nonnegative matrix, says

$$\rho(A) = \inf_{t>0} \max_m \frac{(At)_m}{t_m}.$$

Subtracting this identity from s , and using

$$s - \max_m \frac{(At)_m}{t_m} = \min_m \frac{(Q_\Omega t)_m}{t_m},$$

proves (52). Since Ω is symmetric, its connected components give an orthogonal block decomposition. On each nontrivial component, Perron–Frobenius applied to $sI - Q_\Omega$ supplies a strictly positive eigenvector corresponding to the smallest eigenvalue of that block. Combining these vectors, and using a positive scalar on every isolated vertex, gives the asserted positive maximizer. $\square$

Remark 6.5 (Sharpness and scope). For two columns the comparison estimate is exact whenever their inner product has the extremal negative phase. More explicitly, for $e_1, e_2 > 0$ and $0 \leq \omega \leq \sqrt{e_1 e_2}$, the positive semidefinite Gram

$$\begin{pmatrix} e_1 & -\omega \\ -\omega & e_2 \end{pmatrix}$$

can be realized using one shared output, with private outputs carrying any remaining diagonal energy, and with $\Omega_{1,2} = \omega$. Hence neither the coefficient one in (51) nor the weighted optimization in (52) can be improved from only the diagonal energies and phase-blind pair envelopes.

Normalize the overlap matrix by

$$W_{m,n} := \frac{\Omega_{m,n}}{\sqrt{e_m e_n}}, \quad W_{m,m} := 0. \quad (53)$$

It is symmetric and entrywise nonnegative.

Corollary 6.6 (Spectral-radius and weighted Schur certificates). *One has*

$$\boxed{\beta \geq e_{\min}(1 - \rho(W))_+}. \quad (54)$$

In particular, if there are $t_m > 0$ and $0 < \delta \leq 1$ such that

$$\sum_{n \neq m} W_{m,n} \frac{t_n}{t_m} \leq 1 - \delta \quad (m \in \mathcal{M}), \quad (55)$$

then

$$\beta \geq e_{\min} \delta. \quad (56)$$

Proof. The comparison matrix factors as

$$Q_\Omega = D^{1/2}(I - W)D^{1/2}.$$

Since $\lambda_{\min}(I - W) = 1 - \rho(W)$, congruence gives

$$\lambda_{\min}(Q_\Omega) \geq e_{\min}(1 - \rho(W))$$

whenever the last factor is nonnegative. Combine this with (51). Finally, the upper Collatz–Wielandt formula gives

$$\rho(W) \leq \max_m \sum_{n \neq m} W_{m,n} \frac{t_n}{t_m} \leq 1 - \delta,$$

which proves the last assertion. $\square$

6.4 Degree, codegree, and flatness

Let

$$s_m := |S_m|, \quad c_{m,n} := |S_m \cap S_n|, \quad (57)$$

$$\Delta := \max_m |\{n \neq m : S_m \cap S_n \neq \emptyset\}|, \quad \varrho_m := s_m \max_{y \in S_m} \frac{|a_{y,m}|^2}{e_m}. \quad (58)$$

Every $\varrho_m \geq 1$; equality means that the squared magnitudes in column m are flat on their support. Define the phase-blind codegree matrix

$$W_{m,n}^{\text{sup}} := \begin{cases} c_{m,n} \sqrt{\frac{\varrho_m \varrho_n}{s_m s_n}}, & m \neq n, \\ 0, & m = n. \end{cases} \quad (59)$$

Proposition 6.7 (Low-codegree lower-frame certificate). *The normalized overlap matrix obeys*

$$0 \leq W \leq W^{\text{sup}}$$

entrywise. Consequently,

$$\boxed{\beta \geq e_{\min}(1 - \rho(W^{\text{sup}}))_+}. \quad (60)$$

If

$$s_* := \min_m s_m, \quad c_* := \max(\{c_{m,n} : m \neq n\} \cup \{0\}), \quad \varrho_* := \max_m \varrho_m,$$

then the coarser but explicit estimate is

$$\boxed{\beta \geq e_{\min} \left(1 - \frac{\varrho_* \Delta c_*}{s_*} \right)_+}. \quad (61)$$

Proof. For every common output,

$$\frac{|a_{y,m}|}{\sqrt{e_m}} \frac{|a_{y,n}|}{\sqrt{e_n}} \leq \sqrt{\frac{\varrho_m \varrho_n}{s_m s_n}}.$$

Summing over the $c_{m,n}$ common outputs proves the entrywise bound. Monotonicity of the Perron root for nonnegative matrices gives $\rho(W) \leq \rho(W^{\text{sup}})$, and (60) follows from (54).

Every row of W^{sup} has at most Δ nonzero off-diagonal entries, each at most $\varrho_* c_*/s_*$. Therefore

$$\rho(W^{\text{sup}}) \leq \|W^{\text{sup}}\|_{\ell^\infty \rightarrow \ell^\infty} \leq \frac{\varrho_* \Delta c_*}{s_*},$$

which proves (61). $\square$

Remark 6.8 (Strict margin and claim level). The strict inequality $\rho(W^{\text{sup}}) < 1$, or an equivalent weighted margin, is the substantive hypothesis. Nonempty supports, bounded codegree, and small pair overlap considered without the collision degree do not supply it. All statements in this section are finite-dimensional L0 theorems for the actual matrix B . They begin after the activated-range restriction and make no new claim about the general shorted-operator calculus of TPC-62. An arithmetic L1 use must identify $a_{y,m}$, S_m , and e_m with the literal coherently grouped coefficients at one fixed nonzero shift and in the physical atomic norm. None of the displayed identities proves a polynomial exposure, a strict spectral-radius margin, a parity estimate, or a prime-pair lower bound.

7 Forest exactness, matching certificates, and cycle holonomy

We continue with a finite family of nonzero columns $g_m \in \ell^2(\mathcal{Y})$, but restate every object needed in this section. Inner products are conjugate-linear in the first variable. Let

$$e_m = \|g_m\|^2, \quad v_m = \frac{g_m}{\sqrt{e_m}}, \quad C_{m,n} = \langle v_m, v_n \rangle.$$

Thus $C \succeq 0$ and $C_{m,m} = 1$. The actual support of g_m is

$$S_m = \{y : g_m(y) \neq 0\}.$$

7.1 Exact phase removal on forests

The actual Gram graph $\mathcal{G}_C$ has vertex set $\mathcal{M}$ and an edge $\{m, n\}$ exactly when $C_{m,n} \neq 0$. Let W_C be its nonnegative weighted adjacency matrix:

$$(W_C)_{m,n} := \begin{cases} |C_{m,n}|, & m \neq n, \\ 0, & m = n. \end{cases} \quad (62)$$

Theorem 7.1 (Forest-exact comparison). *If $\mathcal{G}_C$ is a forest, then there is a diagonal unitary matrix U such that*

$$\boxed{U^* C U = I - W_C.} \quad (63)$$

Consequently,

$$\boxed{\lambda_{\min}(C) = 1 - \rho(W_C).} \quad (64)$$

In particular, positivity of the normalized lower-frame constant is equivalent to $\rho(W_C) < 1$.

Proof. Choose a root in every tree component and give it phase $u_m = 1$. Proceed away from the root. If n is a child of m , choose the unit scalar u_n so that

$$\overline{u_m} C_{m,n} u_n = -|C_{m,n}|.$$

There is no cycle, so this recursive prescription cannot impose two inconsistent phases on one vertex. With $U = \text{diag}(u_m)$, every nonzero off-diagonal entry of $U^* C U$ is the negative of its magnitude, proving (63).

Unitary similarity preserves eigenvalues. Since W_C is symmetric and entrywise nonnegative,

$$\lambda_{\min}(I - W_C) = 1 - \lambda_{\max}(W_C) = 1 - \rho(W_C),$$

which proves (64). □

Corollary 7.2 (Support-forest singleton formula). *Suppose the support collision graph, with an edge when $S_m \cap S_n \neq \emptyset$, is a forest, and suppose every colliding pair has codegree one. If $y(m, n)$ is its unique common output, then*

$$|C_{m,n}| = \frac{|g_m(y(m, n))| |g_n(y(m, n))|}{\sqrt{e_m e_n}}. \quad (65)$$

Therefore the normalized lower-frame constant is given exactly by (64), independently of all entry phases.

Proof. With only one common output, the inner product of the two columns consists of one nonzero summand, whose magnitude is (65). The actual Gram graph is a subforest of the support collision graph, so Theorem 7.1 applies. □

Corollary 7.3 (Exact star law). *Suppose one connected component of the normalized Gram has center 0, leaves $1, \dots, d$, and no leaf–leaf edges. Put*

$$\gamma_j := C_{0,j}.$$

The component Gram has eigenvalues

$$1 - \|\gamma\|_2, \quad 1 \text{ with multiplicity } d - 1, \quad 1 + \|\gamma\|_2. \quad (66)$$

Hence its exact lower-frame constant is

$$\boxed{1 - \left(\sum_{j=1}^d |\gamma_j|^2 \right)^{1/2}}. \quad (67)$$

Proof. After the forest gauge, the component is

$$\begin{pmatrix} 1 & -w^* \\ -w & I_d \end{pmatrix}, \quad w = (|\gamma_1|, \dots, |\gamma_d|)^T.$$

The leaf subspace orthogonal to w has eigenvalue one. Restriction to the span of the center and w gives eigenvalues $1 \pm \|w\|_2$. $\square$

Proposition 7.4 (Unnormalized star Schur test). *Let a Hermitian Gram component have the star form*

$$H = \begin{pmatrix} e_0 & c^* \\ c & \text{diag}(e_1, \dots, e_d) \end{pmatrix}, \quad e_j > 0.$$

Then

$$\boxed{H \succ 0 \iff e_0 > \sum_{j=1}^d \frac{|c_j|^2}{e_j}}. \quad (68)$$

Proof. The lower-right diagonal block is positive definite. The assertion is exactly its scalar Schur-complement criterion. $\square$

7.2 Hall deficiency, generic rank, and unique matching

Let $\mathcal{B}$ be the bipartite support graph with left vertex set $\mathcal{M}$, right vertex set $\mathcal{Y}$, and edge $m \sim y$ when $y \in S_m$. For $J \subseteq \mathcal{M}$, write

$$N(J) := \bigcup_{m \in J} S_m.$$

Let $r := |\mathcal{M}|$, and let

$$B = (a_{y,m})_{y \in \mathcal{Y}, m \in \mathcal{M}}, \quad \beta = \lambda_{\min}(B^* B).$$

Theorem 7.5 (Support-rank hierarchy). *The following three statements have different quantifiers.*

1. *If some $J \subseteq \mathcal{M}$ satisfies*

$$|N(J)| < |J|, \quad (69)$$

then $\beta = 0$ for every assignment of values on the declared support.

2. *If*

$$|N(J)| \geq |J| \quad (J \subseteq \mathcal{M}), \quad (70)$$

then a generic assignment of nonzero complex values on the support has full column rank. This conclusion is generic, not a statement about the one literal coefficient array.

3. Suppose there is a subset $\mathcal{Y}_0 \subseteq \mathcal{Y}$, with $|\mathcal{Y}_0| = r$, for which the square induced support graph has a unique perfect matching $\phi : \mathcal{M} \rightarrow \mathcal{Y}_0$. Then every assignment which is nonzero on the declared support has full column rank. If

$$T := B|_{\ell^2(\mathcal{Y}_0)}$$

is the corresponding square row restriction, then

$$\beta \geq \sigma_{\min}(T)^2 \geq \frac{\prod_{m \in \mathcal{M}} |a_{\phi(m), m}|^2}{\|T\|^{2(r-1)}}. \quad (71)$$

Proof. If (69) holds, the columns indexed by J are contained in the coordinate space $\ell^2(N(J))$, whose dimension is strictly smaller than $|J|$. Those columns are linearly dependent.

Under (70), Hall's theorem gives a matching of all columns into r distinct output coordinates. In the determinant of the corresponding square minor, this matching supplies a nonzero monomial in the independent entry variables. The determinant polynomial is therefore not identically zero, so it is nonzero outside a proper algebraic set of fillings. This proves the generic statement.

Under the unique-matching hypothesis, the determinant of T has exactly one nonzero permutation term:

$$\det T = \pm \prod_{m \in \mathcal{M}} a_{\phi(m), m} \neq 0.$$

Thus T , and hence B , has full column rank. Removing output rows can only decrease a Gram, so

$$B^* B \succeq T^* T$$

and $\beta \geq \sigma_{\min}(T)^2$. Finally,

$$|\det T| = \prod_{j=1}^r \sigma_j(T) \leq \|T\|^{r-1} \sigma_{\min}(T),$$

which proves (71). $\square$

Remark 7.6 (Sharpness and asymptotic limitation). The determinant–norm inequality in (71) is sharp: take a diagonal matrix with $r - 1$ diagonal entries equal to $\|T\|$ and one smaller entry. Nevertheless, the product of r matching pivots may be exponentially small in r . Unique matching therefore closes a finite-scale kernel gate, but supplies no polynomial asymptotic lower bound without quantitative pivot and complexity control.

Proposition 7.7 (Private-coordinate peeling). *Suppose a choice of r output rows and an ordering of rows and columns make the selected square matrix triangular with nonzero diagonal. Write*

$$T = D_p(I + N),$$

where D_p is diagonal, N is strictly triangular, and

$$p_{\min} := \min_j |(D_p)_{j,j}| > 0.$$

Then

$$\sigma_{\min}(B) \geq \sigma_{\min}(T) \geq \frac{p_{\min}}{\sum_{k=0}^{r-1} \|N\|^k}. \quad (72)$$

If $\|N\| < 1$, this implies

$$\sigma_{\min}(B) \geq p_{\min}(1 - \|N\|). \quad (73)$$

Proof. Since $N^r = 0$,

$$(I + N)^{-1} = \sum_{k=0}^{r-1} (-N)^k.$$

Therefore

$$\|T^{-1}\| \leq \frac{1}{p_{\min}} \sum_{k=0}^{r-1} \|N\|^k.$$

Taking reciprocals gives the first estimate. If $\|N\| < 1$, bound the finite geometric sum by $(1 - \|N\|)^{-1}$. $\square$

7.3 Cycles retain a gauge-invariant phase

For a graph containing cycles, diagonal column phases cannot remove every edge phase independently. The product of oriented edge phases around a cycle is invariant under diagonal unitary conjugation. The following model calculates the resulting spectral freedom exactly.

Theorem 7.8 (Uniform cycle-holonomy law). *Let $r \geq 3$, $0 \leq a \leq 1/2$, and let C_φ be a Hermitian $r \times r$ matrix with diagonal one, cycle graph C_r , every edge magnitude equal to a , and total oriented edge phase $e^{i\varphi}$. Then*

$$\text{spec}(C_\varphi) = \left\{ 1 + 2a \cos\left(\frac{\varphi + 2\pi k}{r}\right) : k = 0, \dots, r-1 \right\}. \quad (74)$$

Thus the edge magnitudes and support graph do not determine the smallest eigenvalue; the cycle holonomy is additional data.

Proof. Diagonal unitary conjugation can distribute the total phase equally over the r oriented edges. The resulting magnetic cycle is circulant after the twisted boundary condition is absorbed into the Fourier modes. The modes have angles $(\varphi + 2\pi k)/r$, and the two oriented neighbors contribute

$$ae^{i(\varphi+2\pi k)/r} + ae^{-i(\varphi+2\pi k)/r}.$$

Adding the diagonal one gives (74). $\square$

Example 7.9 (Same supports and magnitudes, zero or positive gap). *Fix $r \geq 3$, outputs $y_1, \dots, y_r$, and columns $m_1, \dots, m_r$. For $|\omega| = 1$, define B_ω by*

$$\begin{aligned} a_{y_j, m_j} &= 2^{-1/2} & (1 \leq j \leq r), \\ a_{y_{j+1}, m_j} &= 2^{-1/2} & (1 \leq j < r), \\ a_{y_1, m_r} &= \omega 2^{-1/2}, \end{aligned} \quad (75)$$

with all other entries zero. Every column has norm one and

$$S_{m_j} = \{y_j, y_{j+1}\},$$

where indices are cyclic. Hence every support has size two, every output degree is two, the collision degree is two, every pair codegree is at most one, the flatness is one, and Hall's condition holds. All these data, including every nonzero magnitude, are independent of ω .

There are exactly two perfect matchings, and direct expansion gives

$$\det B_\omega = 2^{-r/2} (1 + (-1)^{r-1} \omega). \quad (76)$$

If $\omega = (-1)^r$, then

$$\lambda_{\min}(B_\omega^* B_\omega) = 0.$$

If $\omega = (-1)^{r-1}$, Theorem 7.8 with $a = 1/2$ gives

$$\lambda_{\min}(B_{\omega}^* B_{\omega}) = 1 - \cos\left(\frac{\pi}{r}\right) > 0. \quad (77)$$

Phases approaching the singular choice make the least eigenvalue arbitrarily small while preserving the complete support and magnitude ledger.

Remark 7.10 (Sharp support-only boundary). The singular member of Example 7.9 has

$$\frac{\varrho_* \Delta c_*}{s_*} = \frac{1 \cdot 2 \cdot 1}{2} = 1.$$

It therefore attains the threshold of the coarse low-codegree certificate. The strict inequality in that certificate cannot be replaced by a nonstrict one. The example also shows that Hall's condition, bounded degree and codegree, flat nonzero magnitudes, and unit diagonal exposure do not determine the literal lower-frame constant.

Remark 7.11 (L0/L1 boundary and nonduplication). The forest gauge, matching hierarchy, determinant bound, and cycle model are finite-dimensional L0 results. They analyze the actual Gram after the activated carrier and represented map have already been identified. They neither restate nor strengthen the master shorting theorem of TPC-62. For an arithmetic L1 specialization, the supports must be those of the literal coherently summed entries at one fixed nonzero shift; potential incidence, generic matrix fillings, and freely chosen cycle phases cannot be substituted for the physical coefficients. No theorem here proves a favorable physical holonomy, a polynomial matching-pivot product, a parity-breaking estimate, or a prime-pair conclusion.

8 Sharp route-separation models

These models are finite-dimensional or coefficient-blind diagnostics. They do not contradict the literal arithmetic theorems.

Example 8.1 (Large candidate exposure, empty terminal support). *Take any number of congruence candidates in (13), but let the terminal prime indicator or a literal mask delete every candidate. Then the actual row energy is zero. An upper census cannot be reversed into activation.*

Example 8.2 (Many cells, no diffuseness). *For $N \geq 2$ and $0 < \varepsilon < 1$, let a unit column have squared magnitudes*

$$1 - \varepsilon, \quad \frac{\varepsilon}{N-1}, \dots, \frac{\varepsilon}{N-1}$$

on N cells. Its support grows with N , while its cell concentration is $\rho = 1 - \varepsilon$. Exposure cardinality alone gives no decay of ρ .

Example 8.3 (Pair codegree one, arbitrary global rank loss). *Let N unit columns all be supported on the same single output. Every distinct pair has codegree one, yet the Gram is the all-ones matrix and has rank one. Subpower pair codegree is not a global lower-frame theorem when the row collision degree is unbounded.*

Example 8.4 (Same support data, different pair geometry). *The normalized columns*

$$2^{-1/2}(1, 1), \quad 2^{-1/2}(1, 1)$$

have a kernel, whereas

$$2^{-1/2}(1, 1), \quad 2^{-1/2}(1, -1)$$

are orthogonal. Exposure, pair codegree, output degrees, and entry magnitudes are identical. The relative phase is missing support data.

Example 8.5 (γ -decoration does not equal reduced codegree). *Let two rows share one reduced cell and let its γ -space have dimension $G \geq 2$. If both decorated vectors are nonzero in all G coordinates, the raw full-key support intersection is G , while the reduced codegree is one. Their normalized overlap can be any number in $[0, 1]$. Thus raw γ -multiplicity and Gram-relevant cell codegree are different objects.*

Example 8.6 (Private-energy bound is sharp). *For $0 \leq \delta \leq 1$, take two normalized columns*

$$v_1 = (\sqrt{\delta}, 0, \sqrt{1-\delta}), \quad v_2 = (0, \sqrt{\delta}, \sqrt{1-\delta}).$$

Each row has private energy δ , and the Gram eigenvalues are δ and $2-\delta$. Hence the private-output lower bound equals the true smallest eigenvalue.

The cycle-holonomy section supplies the strongest combined no-go: even bounded row support, output degree, collision degree, pair codegree, Hall matching, and fixed magnitudes do not determine whether the lower frame is positive.

9 Conclusions, dependency graph, and the next gate

9.1 What advanced

The paper proves three kinds of results.

- (1) **Literal row geometry.** A fixed physical row is a singleton in each residual output fiber. The active normalized row Gram therefore has diagonal exactly one.
- (2) **Arithmetic pair geometry.** Summing the TPC-57 exchange patterns for a fixed pair of distinct rows gives

$$C_{m,n}^{\text{red}}(R) \leq X^{o(1)} \left(1 + \frac{J}{R} \right).$$

At $R \asymp J$, fixed row pairs have subpower reduced codegree even though a single output fiber can retain polynomial degree.

- (3) **Lower-frame support calculus.** Private energy, the absolute comparison matrix, forests, stars, and unique matchings give exact positive certificates. Cycle holonomy proves the sharp boundary of support-only data.

This is a real structural advance from the TPC-65 abstract Gram: the unknown matrix is now a unit-diagonal Gram whose reduced-cell pair codegrees are subpower on the critical block. The full decorated support and the global row collision graph may still be large. This is not yet an endpoint lower bound.

9.2 What remains open

No result here proves:

- that every relevant row is nonempty or has positive private energy;
- a fixed-power diffuseness bound for $\rho_m(R)$;
- a favorable bound for the aggregate collision spectral radius or for cycle holonomies;
- a polynomial lower bound for the physical smallest singular value;
- a native/reassembly signed estimate, parity improvement, or prime-pair lower bound.

The exact matrix/support statements are L0. The fixed- h_0 row singleton, factor exchange, CRT spacing, divisor thinning, and critical pair-codegree bound are L1 on the declared literal carrier. No theorem specializes to $h_0 = 2$ through any property special to that shift, and no theorem reaches L2.

9.3 Route update for TPC-67

The next paper should not seek Möbius randomness in the residual Gram: the row/output gauge removes that sign. The smallest next gate is global geometry on the full collision skeleton, using only the pair thinning that has actually been proved:

- (i) quantify Hall deficiency and uniquely matched minors on the actual critical incidence;
- (ii) bound the nonpeelable collision core, alternating cycles, and cycle-holonomy dimension;
- (iii) combine pair codegree, right degree, and actual row-energy diffuseness through Cauchy–Binet or trace/minor identities.

If these mechanisms yield only exponentially small determinant bounds, that is a route stop for uniform closure but still a precise negative theorem. If they give a polynomially conditioned minor or a strict comparison-matrix margin, the route advances to reassembly. Signed two-affine cancellation remains a separate later door.

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